

FORK INFOSYSTEMS

Karad, Maharashtra

SHELL SCRIPTING BASICS ON WINDOWS

Command Line & Batch (.bat) Scripting — Complete Notes

Topic	Windows Shell Scripting (CMD & Batch Files)
Covers	Command Prompt basics, CMD commands, .bat file creation, execution, automation scripts with examples
Includes	Point-by-point notes, examples, diagrams & a ready-to-use ZIP of sample .bat scripts
Prepared For	Java Full Stack Developer Course Students
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INTRODUCTION TO SHELL SCRIPTING ON WINDOWS

What is a Shell?

A **shell** is a program that takes commands from the keyboard and gives them to the operating system to perform. On Windows, the most common shells are the **Command Prompt (cmd.exe)** and **PowerShell**. This document focuses on the classic Command Prompt and Batch (.bat) scripting, which is widely used for automation, file handling and system administration tasks.

What is Shell Scripting?

Shell scripting means writing a sequence of commands in a text file so that they can be executed together, instead of typing them one by one. On Windows, these scripts are saved with a **.bat** (Batch) or **.cmd** extension and run using the Command Prompt.

Why Learn Shell / Batch Scripting?

- Automate repetitive tasks (file creation, copying, backups)
- Run multiple programs / installations one after another
- Manage files, folders and system settings quickly
- Useful for DevOps, deployment scripts and build automation
- Build small utility tools (shutdown timers, cleanup scripts, etc.)

Shell vs Batch vs PowerShell — Quick Comparison

Feature	CMD / Batch (.bat)	PowerShell (.ps1)
Introduced	Since MS-DOS era	Modern (.NET based)
Syntax	Simple, command based	Object-oriented, advanced
Use Case	Basic automation, quick scripts	Advanced scripting, system admin
File Extension	.bat / .cmd	.ps1
Learning Curve	Easy	Moderate to Advanced

MODULE 01 Command Prompt (CMD) Basics

Opening the Command Prompt

- Press **Windows + R**, type **cmd**, press Enter
- Or search 'Command Prompt' in the Start Menu
- Run as Administrator: Right-click → 'Run as administrator' (needed for system-level commands)

CMD Window Layout (Diagram)

```
+-----+
| C:\Users\Student> <-- Command Prompt (Prompt + Path) |
| |
| > Type your command here and press ENTER |
| |
| Output of the command appears below |
| ..... |
| |
| C:\Users\Student> <-- Ready for next command |
+-----+
```

Essential CMD Commands

Command	Purpose	Example
dir	List files & folders in current directory	dir
cd	Change directory	cd Documents
cd ..	Move one folder up	cd ..
md / mkdir	Make (create) a new directory	md Projects
rd / rmdir	Remove a directory	rd Projects
copy	Copy files	copy a.txt b.txt
move	Move / rename files	move a.txt D:\Backup\
del	Delete a file	del a.txt
type	Display file content	type a.txt
cls	Clear the screen	cls
echo	Display message / enable-disable echo	echo Hello World
exit	Close the command prompt	exit
ipconfig	Show network configuration	ipconfig
ping	Check network connectivity	ping google.com
tasklist	Show running processes	tasklist
taskkill	Kill a running process	taskkill /IM notepad.exe /F

■ **Note:** Use **cd /d D:\Folder** to change drive AND directory in one command.

MODULE 02 Batch (.bat) File Basics

What is a .bat File?

A **.bat (Batch) file** is a plain text file containing a series of CMD commands, executed line by line in order when the file is run. It is one of the simplest ways to automate tasks on Windows.

How to Create a .bat File — Step by Step

- Open **Notepad** (or any text editor like VS Code)
- Type your CMD commands, one per line
- Click **File** → **Save As**
- Set 'Save as type' to '**All Files**'
- Name the file with a **.bat** extension, e.g. **myscript.bat**
- Click **Save**

Example: A Simple .bat File

```
@echo off
echo Hello, Welcome to Fork Infosystems!
pause
```

Explanation of Each Line

Line	Meaning
@echo off	Hides command names from output; only shows results
echo Hello...	Prints the given text on screen
pause	Pauses execution and waits for a key press (so window doesn't close instantly)

How to Execute (Run) a .bat File

- **Double-click** the .bat file in File Explorer — it runs in a CMD window
- Run as **Administrator**: Right-click → 'Run as administrator' (for system-level tasks)
- Run from **Command Prompt**: navigate to the folder using **cd**, then type the file name and press Enter, e.g. **myscript.bat**
- Run via **Run dialog**: Press Win+R, type the full path of the .bat file, press Enter

Execution Flow Diagram

```
[Write Commands] [Save as .bat] [Double-click / Run]
| | |
v v v
Notepad / VS Code ---> myscript.bat ---> CMD opens & executes
each line in order
|
v
[Output shown on screen]
```

MODULE 03 Variables, Conditions & Loops

Variables in Batch Scripting

Variables store values (text or numbers) that can be reused. Use **set** to create a variable and **%variable_name%** to use its value.

```
@echo off
set name=Fork Infosystems
set year=2024
echo Welcome to %name%
echo Batch Year: %year%
pause
```

User Input with set /p

```
@echo off
set /p username=Enter your name:
echo Hello, %username%! Welcome.
pause
```

IF-ELSE Conditions

```
@echo off
set /p num=Enter a number:
if %num% GTR 10 (
echo Number is greater than 10
) else (
echo Number is 10 or less
)
pause
```

FOR Loops

Used to repeat a set of commands multiple times.

```
@echo off
for /L %i in (1,1,5) do (
echo Count: %i
)
pause
```

■ **Note:** In a .bat file, use %%i for loop variables (double %). When typing directly in CMD, use single %i.

IF / FOR — Logic Diagram

IF Condition FOR Loop

```
+-----+ +-----+
| Check Condition | | Start: i = 1 |
+-----+ +-----+
| |
True | False v
+-----v-----+ +-----+ +-----+
| Run Block A| | Run Block | | Is i <= 5 ? |
+-----+ | B | +-----+
+-----+ | Yes | No
v v
Run commands Exit loop
|
v
i = i + 1 (repeat)
```

MODULE 04 File & Folder Automation Examples

1. Create a New File

```
@echo off
type nul > MyFile.txt
echo File created successfully!
pause
```

2. Open an Existing File

```
@echo off
start "" "MyFile.txt"
pause
```

3. Copy a File

```
@echo off
copy MyFile.txt MyFile_Copy.txt
echo File copied successfully!
pause
```

4. Create a New Folder

```
@echo off
mkdir MyNewFolder
echo Folder created successfully!
pause
```

5. Create a File Inside a Folder

```
@echo off
mkdir MyNewFolder
type nul > MyNewFolder\Notes.txt
echo File created inside folder!
pause
```

■ **Note:** All five examples above are also provided as ready-made, working **.bat** files in the attached ZIP package — see Module 6.

MODULE 05 Application & Setup Automation

1. Open / Run an Installed EXE

To run any installed application or setup file, use the **start** command followed by the full path of the .exe file.

```
@echo off
start "" "C:\Path\To\Your\Setup.exe"
pause
```

2. Run Multiple EXE Setups Sequentially

The keyword **start /wait** tells CMD to wait until that program/installer is closed before moving to the next line. This is useful for installing multiple software packages one after another without manual intervention.

```
@echo off
echo Starting Setup 1 ...
start /wait "" "C:\Path\To\Setup1.exe"
echo Setup 1 completed.

echo Starting Setup 2 ...
start /wait "" "C:\Path\To\Setup2.exe"
echo Setup 2 completed.

echo All setups installed successfully!
pause
```

Sequential Execution Diagram

```
[Run Script]
|
v
+-----+ wait until closed +-----+
| start /wait | -----> | Setup1.exe runs |
| Setup1.exe | | (user installs) |
+-----+ +-----+
|
closed v
+-----+
| start /wait |
| Setup2.exe runs |
+-----+
|
closed v
[All Setups Done]
```

■ **Note:** Always replace the placeholder paths (e.g. C:\Path\To\Setup1.exe) with the actual location of your installer files.

MODULE 06 Power Options with Timer (Shutdown / Sleep / Hibernate / Logout)

Key Commands

Action	Command	Description
Shutdown	shutdown /s /t <seconds>	Shuts down PC after given time
Restart	shutdown /r /t <seconds>	Restarts PC after given time
Cancel	shutdown /a	Cancels a pending shutdown/restart
Logout	shutdown /l	Logs off the current user
Hibernate	shutdown /h	Puts PC into hibernation
Sleep	rundll32.exe powrprof.dll,SetSuspendState 0,1,0	Puts PC into sleep mode

Example: Shutdown Timer Script

```
@echo off
set /p secs=Enter time in seconds before shutdown:
shutdown /s /t %secs%
echo Shutdown scheduled in %secs% seconds.
echo To cancel, run: shutdown /a
pause
```

Example: Menu-Driven Power Options Script

The complete menu-driven script (with Shutdown, Restart, Sleep, Hibernate, Logout and Cancel options, all with timers) is included as **08_power_options_with_timer.bat** in the attached ZIP file.

```
@echo off
:MENU
echo 1. Shutdown 2. Restart 3. Sleep
echo 4. Hibernate 5. Logout 6. Cancel 7. Exit
set /p choice=Enter choice:
if %choice%==1 goto SHUTDOWN
if %choice%==6 goto CANCEL
...
:SHUTDOWN
set /p secs=Enter seconds:
shutdown /s /t %secs%
goto MENU
:CANCEL
shutdown /a
goto MENU
```

■ **Note:** Run Command Prompt as **Administrator** for shutdown/hibernate commands to work reliably on all systems.

MODULE 07 Attached ZIP — Ready-to-Use Batch Scripts

A ZIP file named **Fork_Infosystems_BatchScripts.zip** is provided along with this document. It contains the following ready-to-run .bat scripts:

File Name	Purpose
01_create_file.bat	Creates a new empty text file
02_open_file.bat	Opens the created file
03_copy_file.bat	Copies the file to a new file
04_create_folder.bat	Creates a new folder
05_create_file_in_folder.bat	Creates a folder and a file inside it
06_open_exe_setup.bat	Opens / runs an installed .exe setup
07_run_multiple_setups_sequentially.bat	Runs multiple .exe setups one after another
08_power_options_with_timer.bat	Menu-driven Shutdown / Sleep / Hibernate / Logout with timer
README.txt	Instructions for using all the scripts

How to Use the ZIP Package

- Extract (unzip) the folder to any location, e.g. Desktop
- Read **README.txt** for instructions
- Right-click any .bat file → 'Edit' to view/modify in Notepad
- Update placeholder paths (for setup scripts) before running
- Double-click a .bat file to execute it

Best Practices for Batch Scripting

- Always start scripts with **@echo off** for clean output
- Add comments using **REM** or **::** to explain code
- Use **pause** at the end so the window doesn't close instantly
- Test scripts on a sample/dummy folder before using on real data
- Take backups before running scripts that delete or move files
- Use full paths in quotes when paths contain spaces

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